# PegGuard: Stress-Testing Stablecoin Reserves against Redemption Runs

GitHub repository: https://github.com/madhupolluru/IS4228-TeamBulge-PegGuard

## 1. What does it do?

Companies that hold fiat-backed stablecoins as working capital judge whether those coins are safe from the issuer's periodic attestation, which reports what the reserves held on a single date. PegGuard turns that report into a day-by-day simulation of a redemption run under market shocks chosen by the user. It reports whether the issuer can still pay every token $1 within a stated deadline, and what that means for the user's own position. The engine is deterministic, every figure can be traced, and it reproduces the outcome of four historical runs. It is delivered as a working Streamlit application that runs offline.

### 1.1 Core Mechanic

Each run passes through four stages:

1. **Input.** The user selects a reserve portfolio, a list of holdings with asset class, amount, maturity, yield and custodian bank, and can edit any line. Six portfolios are bundled, each labelled by data basis: *Reported* (USDC at 30 Sep 2026 and USDT at 30 Jun 2026, from the issuers' own reports), *Reconstructed* (USDC on 10 Mar 2023) and *Hypothetical* (an SVB-style duration mismatch, a Singapore-dollar stablecoin and a narrow benchmark).
2. **Day 0: shocks.** A rate shock (parallel, with an optional curve twist) reprices bills as zero-coupon bonds, $P = (1+y)^{-T}$, and notes as coupon bonds bought at par. Government money market funds keep a $1.00 NAV unless the shadow NAV falls below $0.995. Gold, crypto and equities fall by chosen percentages, secured loans take a credit loss, and a bank failure freezes cash at all banks or at one named bank, with a recovery rate and a day on which the cash becomes usable again.
3. **Days 1 to N: the run.** Redemption requests arrive on a front-loaded schedule. The issuer pays $1 per token by selling through a liquidity waterfall (cash, repo, money market fund, bills, notes, agency mortgage-backed securities, commercial paper, gold, crypto, other investments, loans), limited by each asset's daily liquidity. The fire-sale discount grows with the square root of the amount sold relative to market depth, and also marks down the holdings that were not sold. Unpaid requests queue first-in, first-out and count as late once they pass the chosen deadline (same day to T+5). Optional panic feedback, after Diamond and Dybvig (1983), scales new requests by $1 + k \times 100 \times (1 - s)$ whenever reserves per token, $s$, fall below $1.
4. **Output.** A verdict (HOLDS, LIQUIDITY STRAIN, BELOW PAR or BREAKS) given as answers to three questions: is there enough backing, were redemptions paid on time, and what happens to the user's position if it is held, redeemed on day 1 (net of the issuer's fee and minimum) or sold on the market. Every figure is traced day by day, losses are attributed to their source, and a one-page risk memo can be downloaded.

**Worked example.** For USDT-style reserves under a full panic (rates +150bp, crypto −40%, gold −10%, other investments −30%, a 15% loan loss and a 50% run over five days, with panic feedback) and a same-day deadline, reserves fall to $0.45 per token for the holders left at the end, $35.4bn of requests (19%) are paid late, and a $30m position held to the end loses $16.4m. Allowing the five business days MAS expects cuts late payments to $21.2bn but lowers backing for those who stay to $0.40, because paying early redeemers $1 from short reserves moves the loss onto everyone else.

### 1.2 Analyses Built on the Same Engine

- **Breaking-point map.** 441 runs over rate shocks of 0 to 500bp and run sizes of 0 to 100%, showing the largest run each portfolio survives.
- **Assumption analysis.** A sensitivity table, a tornado chart that changes one judgement-call assumption at a time, and a seeded 300-run Monte Carlo that varies them all together.
- **Treasury Planner.** Applies five standard crises to a treasury of several coins and returns a safe cap for each coin (loss budget ÷ worst haircut).
- **Issuer profile and validation.** Checks reserves against the eligible-asset rules of the US GENIUS Act, the MAS stablecoin framework, EU MiCA and the Hong Kong Stablecoins Ordinance, and replays four historical runs (Section 3.4).

### 1.3 Scope and Deliberate Exclusions

The build was scoped to one reliable core mechanic in about four weeks. We deliberately left out an order-book or arbitrage model of secondary markets, machine-learning price prediction, user accounts, issuer capital injections or

central-bank support during a run, stochastic interest rates, and on-chain proof of reserves. Each would either add assumptions that a risk committee cannot audit or add features that do not change the core answer. Live market data (DefiLlama) is optional, with a cached fallback, so the application runs entirely offline.

# 2. Who is it for?

**Primary user.** The head of treasury at a Singapore Major Payment Institution that settles business-to-business cross-border payouts to Indonesia, the Philippines and Vietnam around the clock. To settle outside banking hours, the firm pre-funds about US$95m of stablecoin float: US$60m in USDC, US$30m in USDT and US$5m in a Singapore-dollar stablecoin. In our demonstration this illustrative composite persona is called "Mei"; she is not a real individual.

**Pain point.** Her risk committee requires (a) an exposure limit for each stablecoin and (b) a contingency plan for a depeg, and MAS expects stablecoins to be redeemed at par within five business days. The only evidence she has is the issuer's monthly attestation or quarterly report, which shows the reserves on one date and is published weeks later, plus letter-grade ratings. Neither answers the committee's actual question: *under what conditions does this coin stop paying $1 within five days, and how much of it can we safely hold?* The risk is concrete: on 11 March 2023, USDC traded at about $0.88 while redemptions were closed for the weekend (CNBC, 2023).

**What PegGuard gives her.** In the Treasury Planner, her US$95m treasury with a 50bp loss limit (US$0.475m) faces a worst-case loss of US$16.4m, from USDT under a full panic. The planner therefore caps her holdings at about US$9.2m of USDC, US$0.9m of USDT and US$2.2m of the Singapore-dollar coin, giving her a documented, auditable basis for each limit.

**Secondary users.** Exchange and custodian risk teams that set collateral haircuts and listing limits on stablecoins.

# 3. Why does it need to exist?

### 3.1 Static Disclosure for a Dynamic Risk

USDT and USDC alone back about $257bn of tokens: $183.6bn for Tether at 30 Jun 2026 (BDO, 2026) and $73.9bn for USDC at 30 Sep 2026 (Circle, 2026). Yet holders judge them from point-in-time composition reports. A backing ratio of roughly 100% hides failure modes that differ from coin to coin, as PegGuard's own results show:

| Portfolio | Where it is weak | PegGuard result (lowest reserves per token) |
|---|---|---|
| USDC-style (Sep 2026) | Concentration of cash in banks | Holds in four of five standard crises; $0.95 if 30% of bank cash is frozen |
| USDT-style (Q2 2026) | Gold, crypto and loans; slow liquidity | Day-1 liquidity covers 48% of tokens; $0.96 in a crypto crash; $0.45 in a full panic |
| SVB-style (hypothetical) | Long-dated Treasuries | $0.81 after a +300bp rate shock |

Standard scenarios as defined in Appendix A; same-day redemption deadline.

### 3.2 Regulation Defines Safe Reserves but Does Not Stress-Test Them

The GENIUS Act (2025), the MAS single-currency stablecoin framework (2023), MiCA (2024) and the Hong Kong Stablecoins Ordinance (2025) list eligible reserve assets and require full backing and prompt redemption at par. They define what a reserve may hold, not how it behaves in a run. Companies also still hold coins whose reserves fall partly outside these lists: by PegGuard's check, only 71% of USDT-style reserves are GENIUS-eligible (75% under MAS), against 100% for USDC.

### 3.3 Existing Tools Look Backwards or Give a Score

Dashboards such as DefiLlama report supply and price after the fact. Rating products, including S&P's Stablecoin Stability Assessment and Bluechip's grades, assign a score. None lets a treasury run *its own* scenario against *its own* redemption deadline and position, and produce a result its risk committee can audit.

### 3.4 Why This Approach Is Appropriate

- **Deterministic and auditable.** A risk committee must be able to trace every number, and there are too few real depegs to train a predictive model. A transparent scenario engine with every assumption visible and editable fits both constraints.

- **Validated against history.** Rebuilt from the last public report before each event, the engine reproduces the outcome of all four historical runs tested: USDT in May 2022, USDC on the SVB weekend, Silicon Valley Bank itself, and BUSD in February 2023 (Appendix B). At the market's price low it is consistently about 3.5 cents too optimistic. We report this gap as a calibrated fear premium rather than hiding it.
- **Anchored and tested assumptions.** Liquidity and price-impact parameters are anchored on market data, for example about $219bn of T-bills traded per day (FINRA, 2026), and on stress spreads from March 2020. The tornado chart and Monte Carlo show which assumptions change the verdict.
- **Established practice and course methods.** The tool applies banks' liquidity stress tests and liquidity ladders to stablecoins, using bond pricing (Week 3), systematic and idiosyncratic risk (Week 5) and bank runs (Weeks 9 and 10). Eighty-two automated tests run on every push. They cover bond prices checked by hand, property tests on random portfolios, the four back-tests, and rendering of every page.

# 4. How could it generate revenue?

We propose a freemium software-as-a-service model, with a data licence for institutions:

| Tier | Customer | What they pay for | Indicative price |
|---|---|---|---|
| Free | Anyone | Public scorecard of major stablecoins under standard scenarios (lead generation) | US$0 |
| Treasury Pro | Payment institutions, fintechs, crypto-native companies | Own portfolios and scenarios, exposure-limit reports and memos for risk committees, alerts when a new attestation moves a coin's breaking point | US$500–1,500 per month |
| Enterprise / data licence | Exchanges, custodians, lenders | API of stress metrics per coin (breaking point, day-1 liquidity, eligible backing) for collateral haircuts and listing limits | US$25,000–100,000 per year |

**Why customers would pay.** The value is avoided loss and documented compliance. A 2% depeg on a US$30m position costs US$0.6m, while a year of Treasury Pro costs US$6,000 to US$18,000, or 1% to 3% of that single loss. Licensed payment institutions in Singapore form a short, identifiable list, and MAS reporting expectations give them a compliance reason to buy.

**Principal risks.** Data goes stale because issuers publish monthly or quarterly; we would mitigate this by ingesting new reports automatically. There is also a conflict of interest if issuers pay for favourable results, so the methodology stays public and we would never accept issuer fees for scores.

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# Appendix A. Standard Scenario Results

Verdict and lowest reserves per token for each bundled portfolio under the five standard crises, with a same-day redemption deadline.

| Portfolio | Calm, 5% run | Rates +300bp, 20% run | Crypto crash, 30% run | Bank failure, 30% frozen | Full panic, 50% run |
|---|---|---|---|---|---|
| USDC-style (Sep 2026) | HOLDS $1.00 | HOLDS $1.00 | HOLDS $1.00 | BELOW PAR $0.95 | HOLDS $1.00 |
| USDT-style (Q2 2026) | HOLDS $1.00 | HOLDS $1.00 | BELOW PAR $0.96 | HOLDS $1.00 | BREAKS $0.45 |
| USDC, 10 Mar 2023 (reconstructed) | HOLDS $1.00 | HOLDS $1.00 | HOLDS $1.00 | BELOW PAR $0.91 | BELOW PAR $1.00 |
| SVB-style (hypothetical) | HOLDS $1.00 | BELOW PAR $0.81 | BELOW PAR $0.99 | BELOW PAR $0.96 | BREAKS $0.00 |
| Singapore-dollar coin (illustrative) | HOLDS $1.00 | HOLDS $1.00 | HOLDS $1.00 | BELOW PAR $0.79 | HOLDS $1.00 |
| Narrow benchmark (hypothetical) | HOLDS $1.00 | HOLDS $1.00 | HOLDS $1.00 | BELOW PAR $0.97 | HOLDS $1.00 |

Values below $1.00 are marked-to-market reserves per remaining token at the low point; $1.00 means every redemption was paid at par. A BELOW PAR verdict shown as $1.00 reflects rounding (USDC 2023 under a full panic: $0.9987). Reproduce any cell with python scripts/run_scenario.py --preset <id> --scenario "<crisis>".

# Appendix B. Historical Back-Tests

Each event was rebuilt from the last public reserve report before it and driven by the redemptions actually observed.

| Event | What happened | PegGuard | Price low, observed vs model |
|---|---|---|---|
| USDT, May 2022 (Terra collapse) | About $7bn redeemed in 48 hours; every redemption paid | Pays everyone on time | $0.950 vs $0.985 |
| USDC, March 2023 (SVB weekend) | $3.3bn frozen at SVB, redemptions closed; par once banks reopened | Holds; paid in full when banks reopen | $0.877 vs $0.912 |
| Silicon Valley Bank, March 2023 | $42bn withdrawn on day 1; closed before it could pay | Cannot pay on day 1 | n/a |
| BUSD, February 2023 | Supply fell from $16.2bn to $7.6bn in seven weeks; all paid | Pays everyone | n/a |

All four outcomes match. Where an observed market low exists, the model is about 3.5 cents (350bp) too high in both cases; we report this as a fear premium that users can add back. Reproduce with python scripts/run_scenario.py --backtest.

# Appendix C. Implementation

| Path | Contents |
|---|---|
| pegguard/ | The model, with no user-interface code: stress engine, bond pricing, analyses, back-tests, risk memo |
| app.py, app_pages/ | Streamlit application, one file per page: Stress Lab, Dashboard, Issuer Profile, Treasury Planner, Validation, Methodology |
| data/ | Bundled reserve portfolios with their sources, and an offline market-data snapshot |
| tests/ | 82 automated tests (pytest), run by continuous integration on every push |
| scripts/ | Command-line runner for any scenario and the back-tests; data-refresh script |

Setup: pip install -r requirements.txt, then streamlit run app.py. No API keys or secrets are required. Educational prototype; not investment or legal advice.